Name: ____________________

STATISTICS, (FREQUENCY TABLES AND GROUPED DATA)

Date: ______________

LO: I can calculate averages and other statistics from frequency tables and grouped frequency tables.

Averages from frequency tables

A **frequency table** shows how often each value (or group of values) occurs. For grouped data, use the **midpoint** of each class to estimate the mean.

Strategy: Mean = Σfx ÷ Σf. For grouped data, x is the class midpoint. The modal class has the highest frequency.

Quick examples

●	Mean = Σfx ÷ Σf	total of fx divided by total frequency
●	Midpoint of 10–20 is 15	(10 + 20) ÷ 2
●	Modal class = class with highest frequency	not the midpoint
●	Mean of grouped data uses class boundaries	use midpoints
	You can find the exact mean	

Key idea

For grouped data: estimate mean = sum of (frequency × midpoint) divided by total frequency.

Estimated mean = Σfx ÷ Σf

x = midpoint of each class; f = frequency

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 – ungrouped frequency table

Scores: 1(f=3), 2(f=5), 3(f=7), 4(f=4), 5(f=1). Find the mean.

Σfx = 1×3 + 2×5 + 3×7 + 4×4 + 5×1 = 3+10+21+16+5 = 55

Σf = 3+5+7+4+1 = 20

Mean = 55 ÷ 20 = 2.75

Mean = 2.75

Multiply each value by its frequency, sum, then divide by total frequency.

Example 2 – grouped data mean

Heights (cm): 140–150 (f=4), 150–160 (f=10), 160–170 (f=8), 170–180 (f=3). Estimate the mean.

Midpoints: 145, 155, 165, 175

Σfx = 145×4 + 155×10 + 165×8 + 175×3 = 580+1550+1320+525 = 3975

Mean = 3975 ÷ 25 = 159

Estimated mean = 159 cm

Use midpoints because we do not know exact values within groups.

Example 3 – modal class and median class

Classes: 0–10 (f=5), 10–20 (f=12), 20–30 (f=8), 30–40 (f=5). Find the modal class and the class containing the median.

Modal class: 10–20 (highest frequency = 12)

Total n = 30; median at $^{30}/_{2}$ = 15th value

CF: 5, 17, ... The 15th value is in the 10–20 class

Modal class: 10–20; Median class: 10–20

Example 4 – range from grouped data

Smallest possible value: 0 (from class 0–10). Largest possible: 40 (from class 30–40). Find the estimated range.

Minimum could be anywhere in 0–10

Maximum could be anywhere in 30–40

Estimated range = 40 - 0 = 40 (maximum possible)

Estimated range = 40

From grouped data, we can only estimate the range using class boundaries.

YOUR TURN – PRACTICE (deliberate practice)

1. Mean from ungrouped frequency tables

Calculate the mean.

a) x: 1,2,3,4; f: 5,8,4,3

b) x: 0,1,2,3,4; f: 2,6,9,7,1

c) x: 5,6,7,8; f: 4,10,8,3

d) x: 10,20,30; f: 7,5,8

2. Estimated mean (grouped)

Find the estimated mean using midpoints.

a) 0–10 (f=6), 10–20 (f=10), 20–30 (f=4)

b) 0–5 (f=3), 5–10 (f=8), 10–15 (f=9)

c) 50–60 (f=5), 60–70 (f=12), 70–80 (f=8), 80–90 (f=5)

d) 100–120 (f=4), 120–140 (f=7), 140–160 (f=9)

3. Modal class and median class

Identify the modal class and the class containing the median.

a) 0–10(f=4), 10–20(f=11), 20–30(f=5)

b) 0–5(f=2), 5–10(f=8), 10–15(f=6), 15–20(f=4)

c) 20–30(f=7), 30–40(f=13), 40–50(f=10)

d) 0–100(f=5), 100–200(f=15), 200–300(f=10)

4. Median and mode from ungrouped

Find the median and mode.

a) x: 1,2,3,4,5; f: 2,5,8,3,2

b) x: 0,1,2,3; f: 4,6,6,4

c) x: 10,11,12,13,14; f: 1,3,7,5,4

d) x: 2,4,6,8; f: 5,10,3,2

5. Mixed frequency table problems

Answer each question.

c) Classes: 0–20(f=8), 20–40(f=12)

d) Modal class for: 1–3(f=4), 3–

e) Estimated mean if all in class 10

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) Mean from a frequency table = Σfx ÷ Σf ☐

Correct answer: __________

Reason: _________________

b) For grouped data, use the class boundaries (not midpoints) for the mean ☐

Correct answer: __________

Reason:

c) The modal class is the class with the highest frequency ☐

Correct answer: __________

Reason: _________________

d) You can find the exact mean from grouped data ☐

Correct answer: __________

Reason: _________________

e) The midpoint of a class 10–20 is 15 ☐

Correct answer: __________

Reason: _________________

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A grouped frequency table shows times (minutes): 0–5 (f=3), 5–10 (f=7), 10–15 (f=**k**), 15–20 (f=4). The estimated mean is 10.5 minutes. Find the value of k.

Expression: ____________________________

Simplified: ____________________________

b) Two datasets are summarised. Set A: n=20, Σfx=140. Set B: n=30, Σfx=240. Find the mean of each set. If the datasets are combined, find the overall mean. Explain why this is not the average of the two means.

Expression: ____________________________

Simplified: ____________________________

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

WELL DONE!